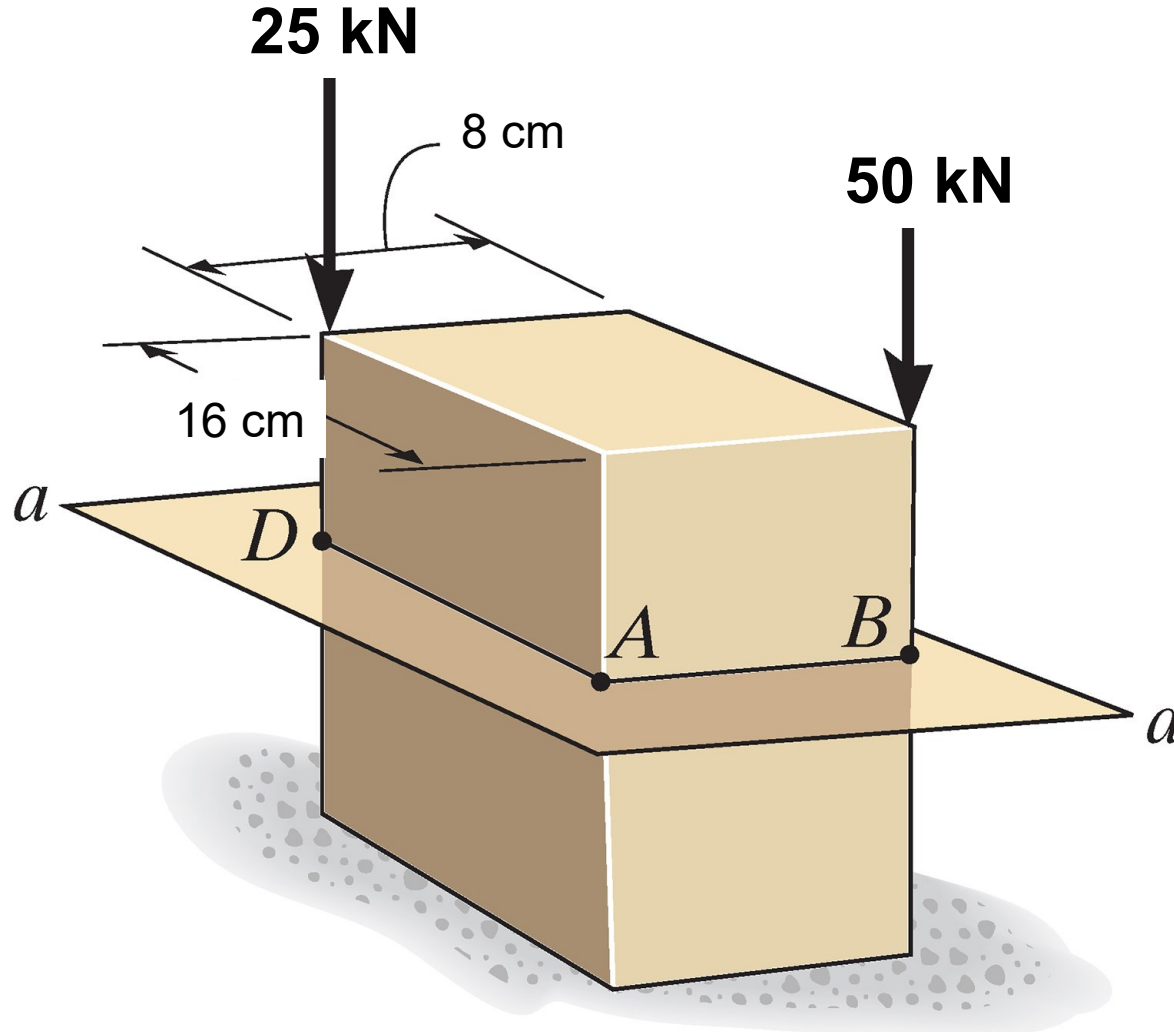


1. Determine the normal stress developed at points A and B . Also, specify which type of the normal stress is (tensile or compressive). (15')



Solutions to Assignment #1:

Step 1: Internal Loadings

Section the member along the ABD plane, where the stress is to be determined. The xyz coordinate system is established with origin at the centroid point of ABD .

The resultant internal normal force and bending moment can be obtained as:

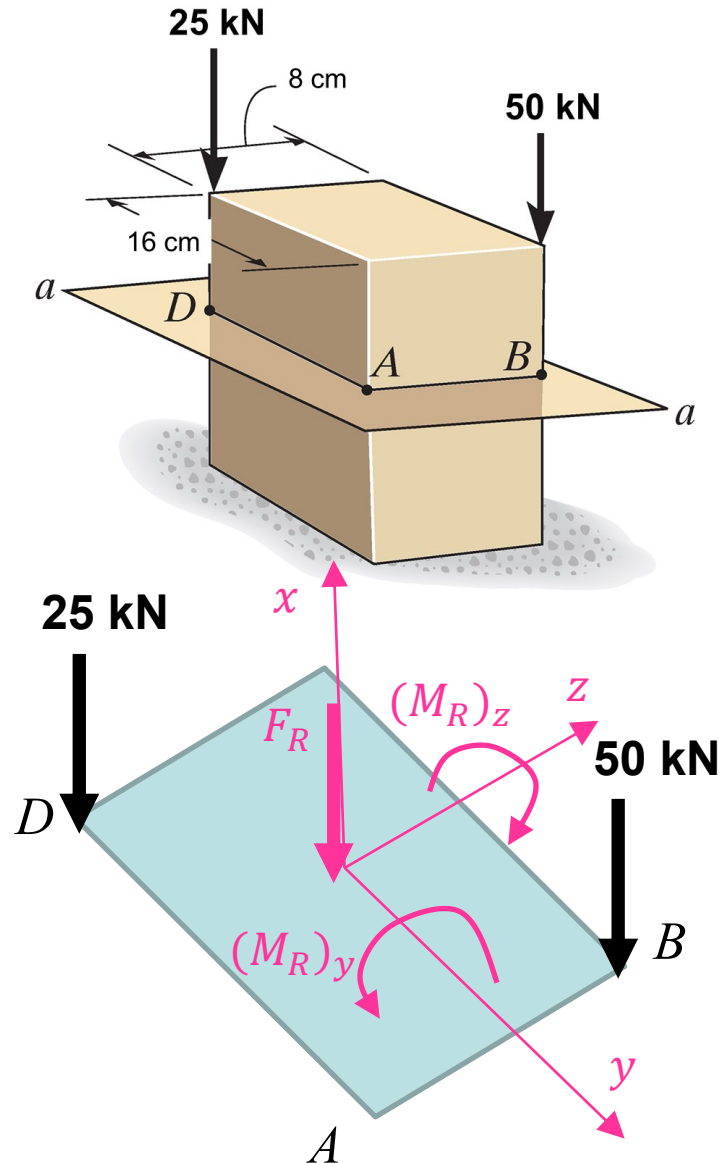
- Normal force in x -direction:

$$F_R = \sum F_x = -25 - 50 = -75 \text{ kN}$$

- Bending moment components:

$$\begin{aligned}(M_R)_y &= \sum M_y = 25000 \times 0.04 - 50000 \times 0.04 \\ &= -1000 \text{ N} \cdot \text{m}\end{aligned}$$

$$\begin{aligned}(M_R)_z &= \sum M_z = 50000 \times 0.08 - 25000 \times 0.08 \\ &= 2000 \text{ N} \cdot \text{m}\end{aligned}$$



Solutions to Assignment #1, cont'd:

Step 2: Cross-section Property

$$A = 0.16 \times 0.08 = 0.0128 \text{ m}^2$$

$$I_y = \frac{1}{12} \times 0.16 \times 0.08^3 = 6.8267 \times 10^{-6} \text{ m}^4$$

$$I_z = \frac{1}{12} \times 0.08 \times 0.16^3 = 2.7307 \times 10^{-5} \text{ m}^4$$

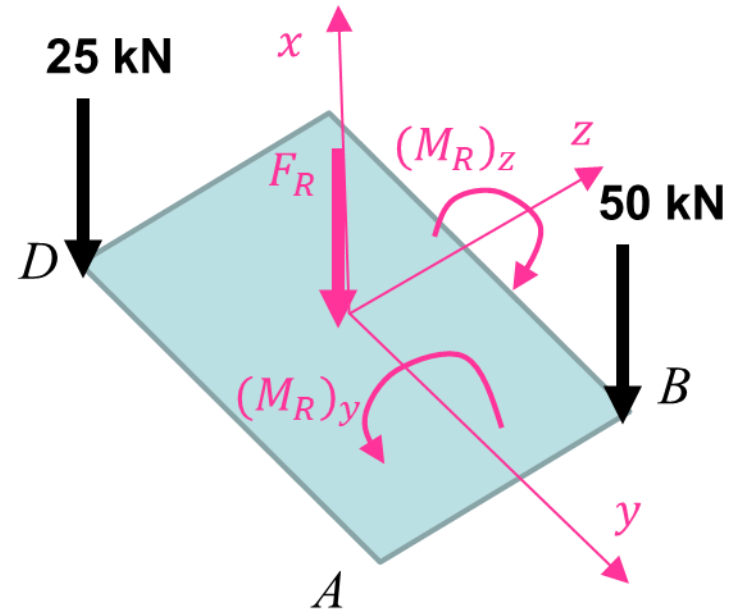
Step 3: Stress Components

The normal stress developed is the combination of axial and bending stress.

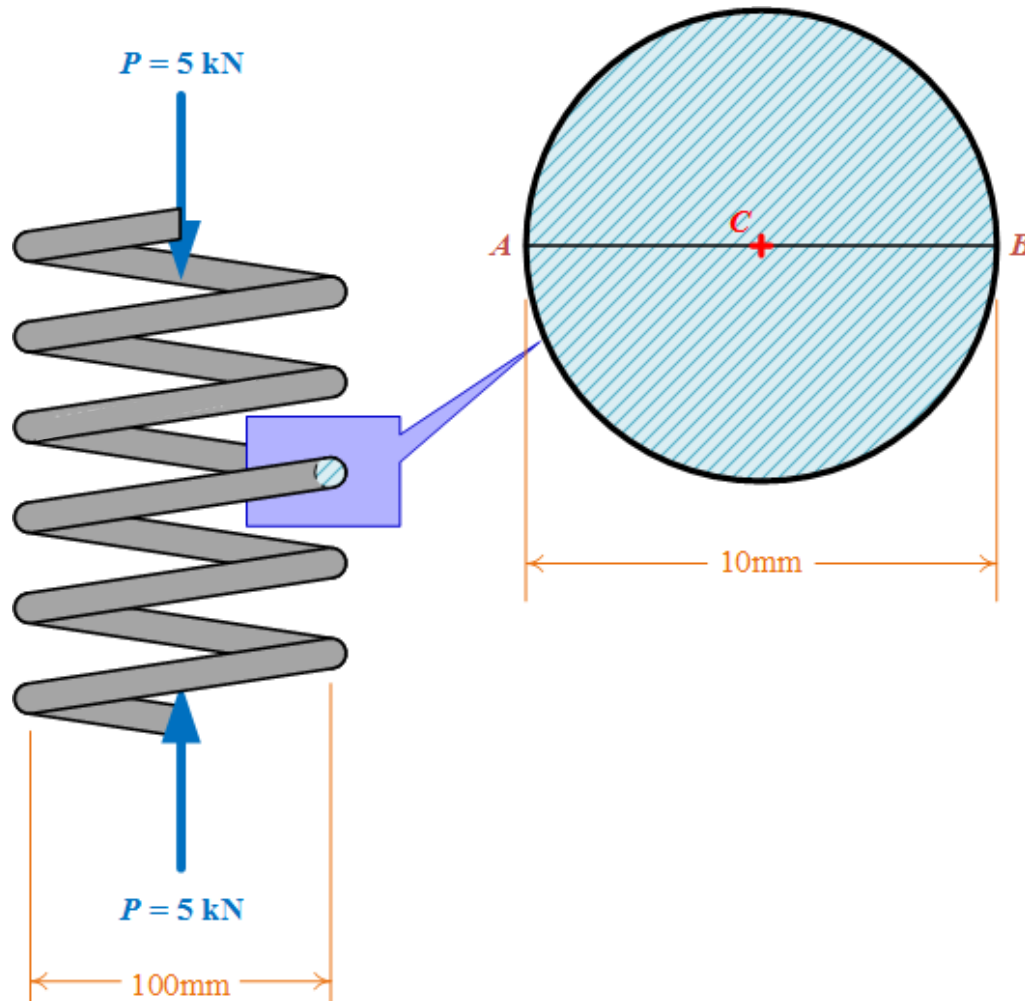
$$\sigma = \frac{F}{A} - \frac{(M_R)_z \cdot y}{I_z} + \frac{(M_R)_y \cdot z}{I_y}$$

$$\text{Thus, } \sigma_A = \frac{-75000}{0.0128} - \frac{2000 \times 0.08}{2.7307 \times 10^{-5}} + \frac{(-1000) \times (-0.04)}{6.8267 \times 10^{-6}} = 5.86 \text{ MPa (tensile) (Ans.)}$$

$$\sigma_B = \frac{-75000}{0.0128} - \frac{2000 \times 0.08}{2.7307 \times 10^{-5}} + \frac{-1000 \times 0.04}{6.8267 \times 10^{-6}} = -17.578 \text{ MPa (compressive) (Ans.)}$$



2. The helical spring shown is axially loaded with a compression force $P = 5 \text{ kN}$. The mean diameter of the spring is 100 mm and the wire used is 10 mm as indicated in the figure. Determine the shear stress at point A and B . (15')



Solutions to Assignment #2

Equivalent Loading (torque):

$$T = 0.05P = 0.05 \times 5000 = 250 \text{ N} \cdot \text{m}$$

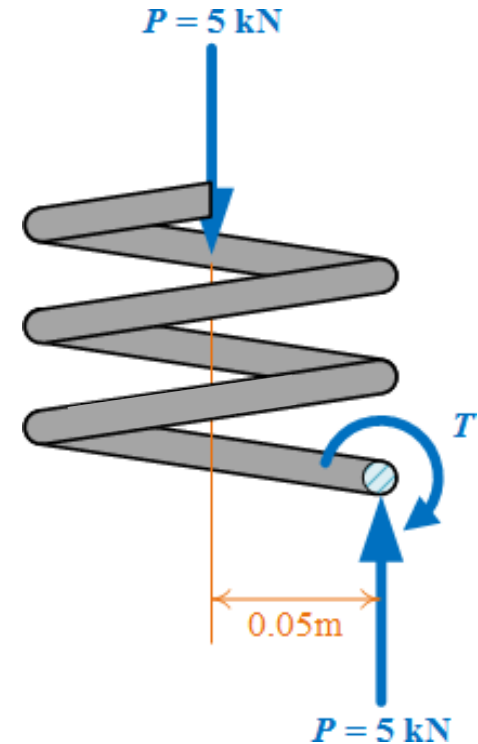
Shear stress due to **direct force**

$$\tau_1 = \frac{P}{A} = \frac{5000}{\pi \times 0.005^2} = 63.662 \text{ MPa}$$

Shear stress due to **torque**

$$\tau_2 = \frac{Tr}{J} = \frac{250 \times 0.005}{\pi/2 \times 0.005^4} = 1273.240 \text{ MPa}$$

The shear stress developed is the combined effect due to direct force and torsion.



Solutions to Assignment #2, cont'd

Shear stress at A (maximum shear stress):

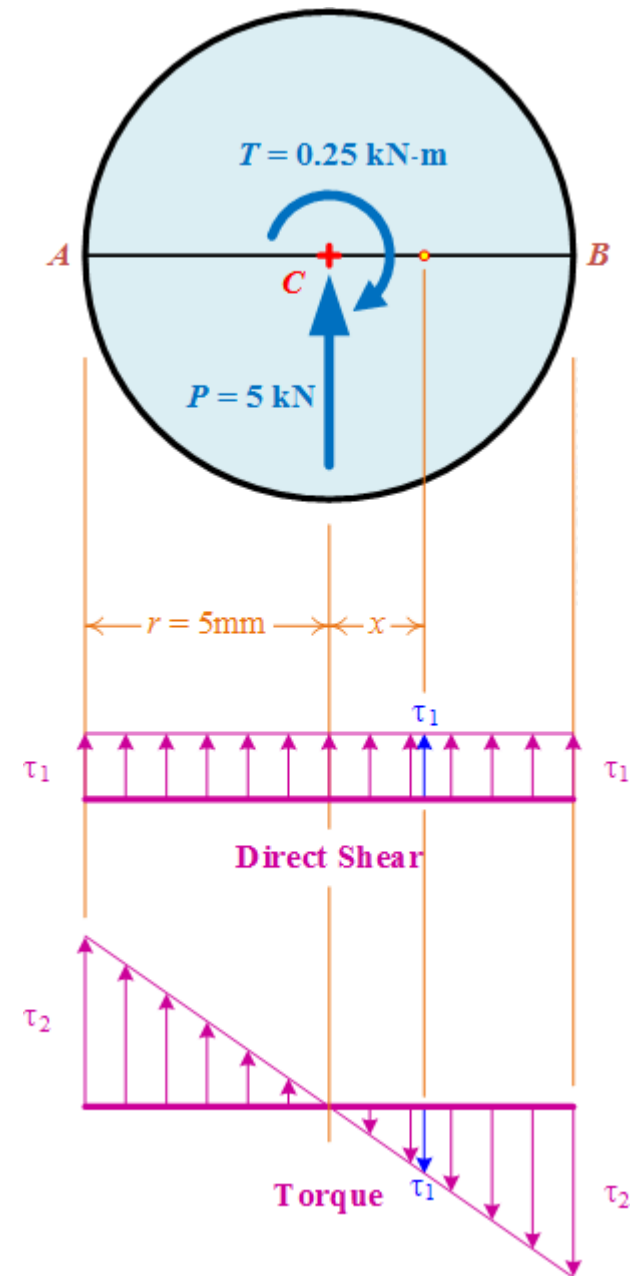
$$\tau_A = \tau_1 + \tau_2$$

$$\tau_A = 63.662 + 1273.240 = 1336.90 \text{ MPa} \quad (\text{Ans.})$$

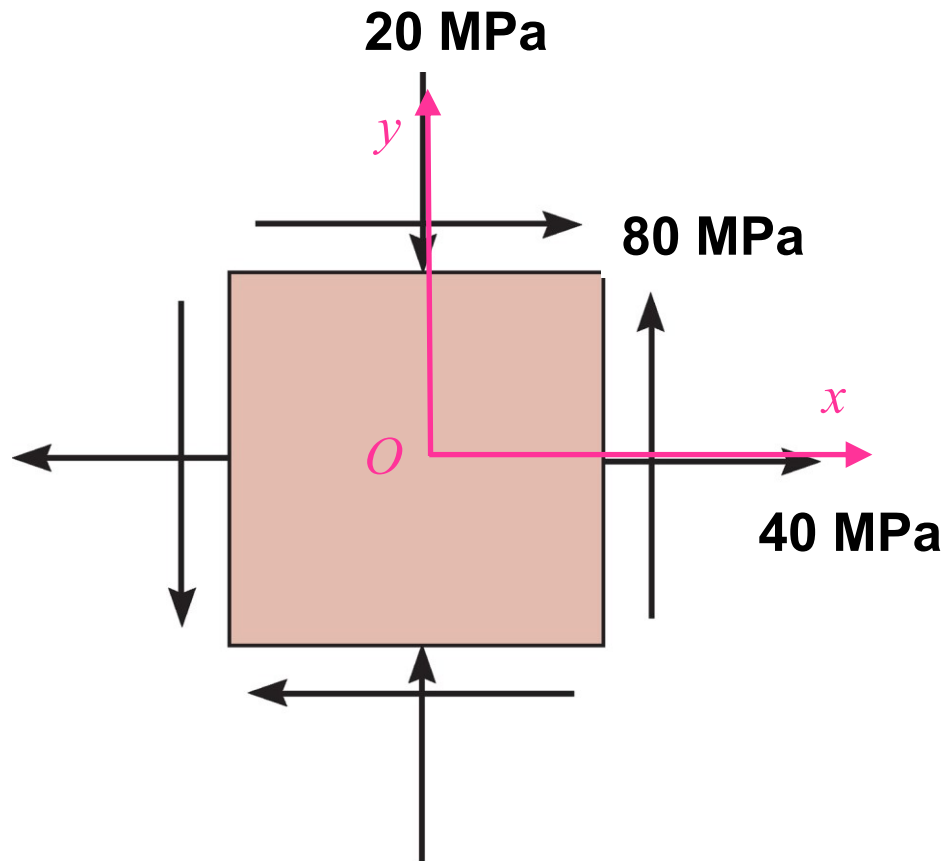
Shear stress at B :

$$\tau_B = \tau_2 - \tau_1$$

$$\tau_B = 1273.240 - 63.662 = 1209.58 \text{ MPa} \quad (\text{Ans.})$$



3. Using the Mohr's circle, determine the principal stresses, the maximum in-plane shear stress, and average normal stress. Specify the orientation of the element in each case. (10' points)

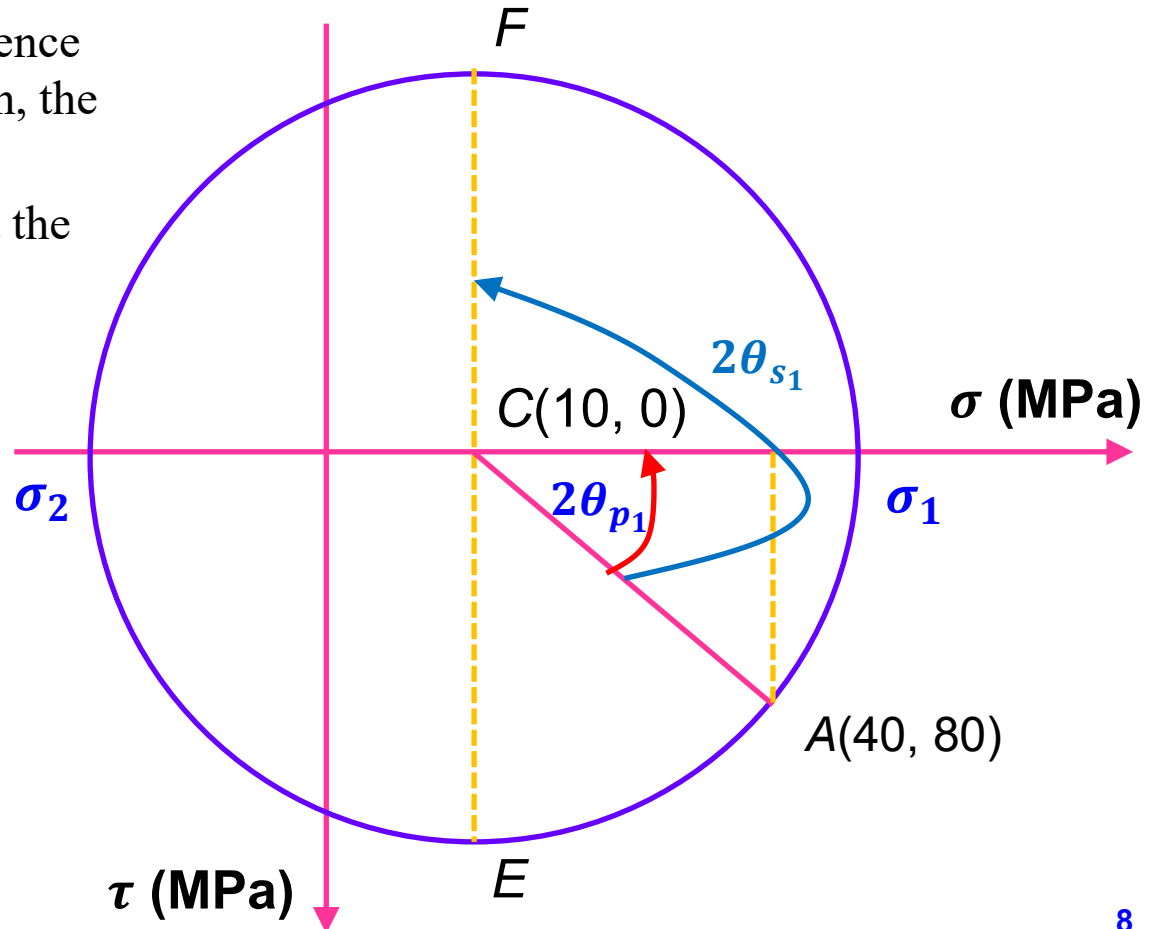


Solutions to Assignment #3

Step 1: Construction of the Mohr's Circle

Given $\sigma_x = 40$ MPa, $\sigma_y = -20$ MPa, $\tau_{xy} = 80$ MPa, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 10$ MPa

The point $C(10, 0)$ and the reference point $A(40, 80)$ are labeled. Then, the radius of the circle is $R = \sqrt{(40 - 10)^2 + 80^2} = 85.4$, and the circle is constructed.



Solutions to Assignment #3, cont'd

Step 2: Principal Stresses

The principal stresses are the two intersections of the Mohr's circle with the σ -axis

$$\sigma_1 = 10 + 85.4 = 95.4 \text{ MPa} \quad (\text{Ans.})$$

$$\sigma_2 = 10 - 85.4 = -75.4 \text{ MPa} \quad (\text{Ans.})$$

From the circle, the counterclockwise angle $2\theta_{p_1}$, measured from CA to the $+\sigma$ axis, is

$$2\theta_{p_1} = \tan^{-1} \left(\frac{80}{40 - 10} \right) = 69.4^\circ$$

$$\text{Thus, } \theta_{p_1} = 34.7^\circ \quad (\text{Ans.})$$

The counterclockwise rotation defines the direction of the x' -axis and σ_1 .

Solutions to Assignment #3, cont'd

Step 3: Maximum In-Plane Shear Stress

The maximum in-plane shear stresses are the upper and lower points on the Mohr's circle:

$$\tau_{\max}^{\text{in-plane}} = R = 85.4 \text{ MPa} \quad (\text{Ans.})$$

The average normal stress is 10 MPa (the σ -axis value of the center point).

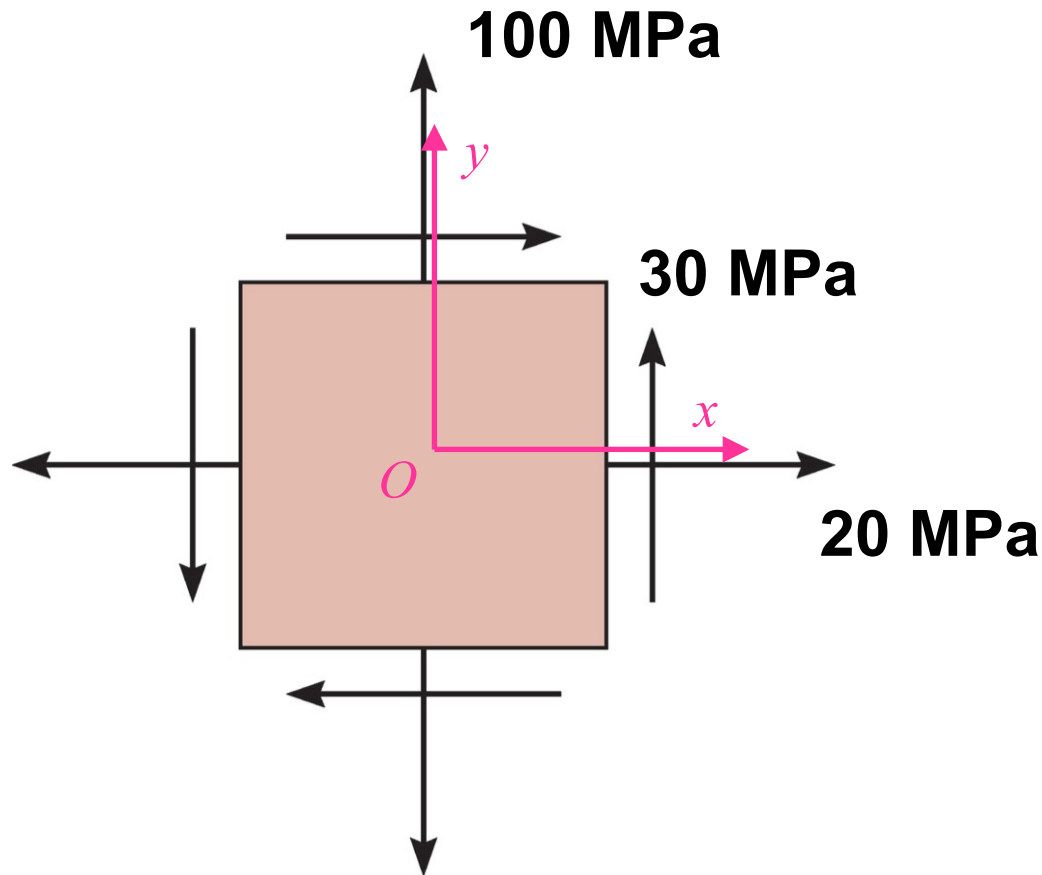
From the circle, the counterclockwise angle $2\theta_{s_1}$, measured from CA to the CF , is

$$2\theta_{s_1} = 2\theta_{p_1} + 90^\circ = 159.4^\circ$$

$$\text{Thus, } \theta_{s_1} = 79.7^\circ \quad (\text{Ans.})$$

The counterclockwise rotation defines the direction of the x' -axis and the negative value of maximum in-plane shear stress.

4. Due to an applied loading, an element at a point on a machine shaft is subjected to the state of plane stress shown below. Determine the principal stresses and the absolute maximum shear stress at the point. (15' points)

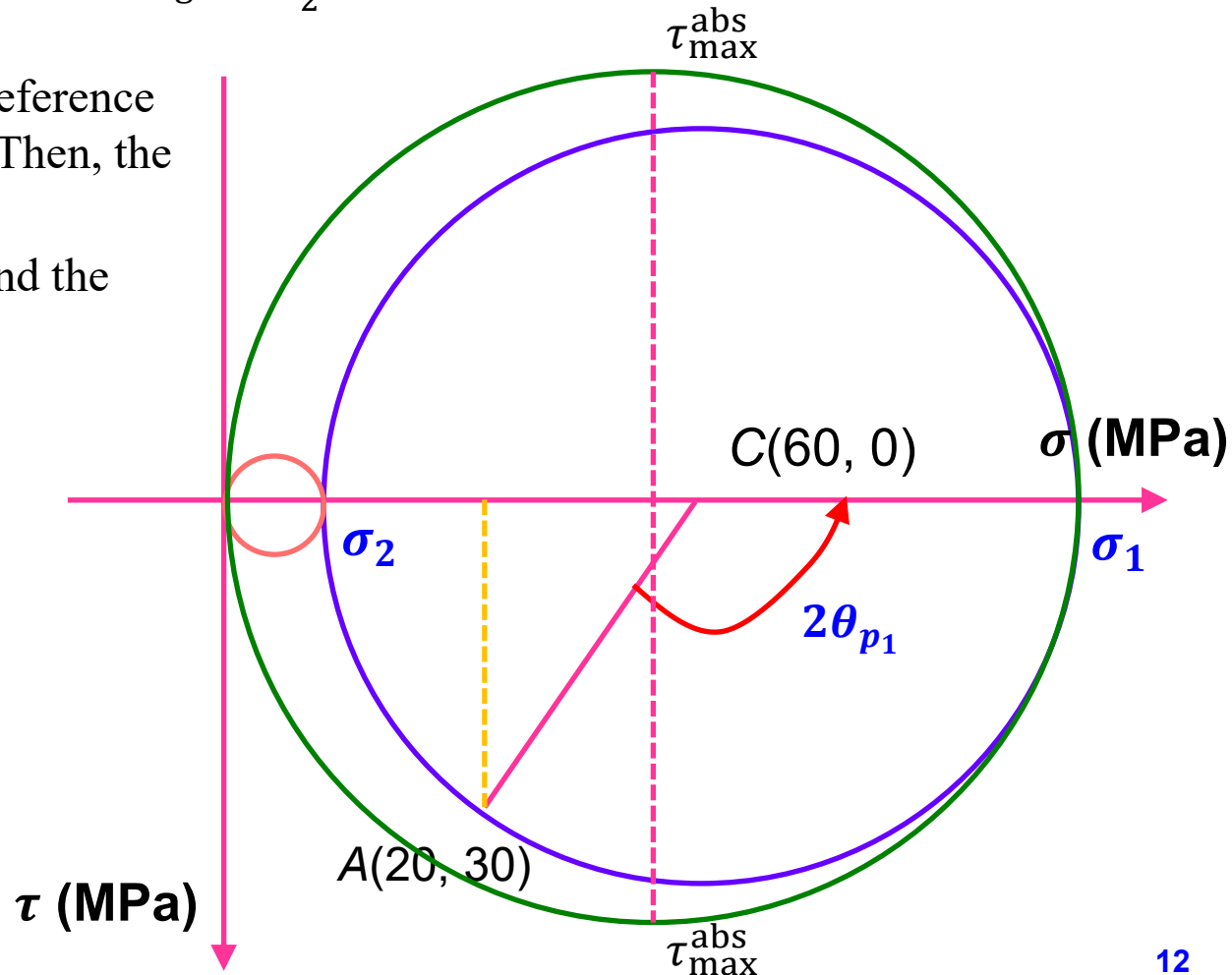


Solutions to Assignment #4

Step 1: Construction of the Mohr's Circle

Given $\sigma_x = 20$ MPa, $\sigma_y = 100$ MPa, $\tau_{xy} = 30$ MPa, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = 60$ MPa

The point $C(60, 0)$ and the reference point $A(20, 30)$ are labeled. Then, the radius of the circle is $R = \sqrt{(60 - 20)^2 + 30^2} = 50$, and the circle is constructed.



Solutions to Assignment #4, cont'd

Step 2: Principal Stresses

The principal stresses are the two intersections of the Mohr's circle with the σ -axis

$$\sigma_1 = 60 + 50 = 110 \text{ MPa} \quad \textbf{(Ans.)}$$

$$\sigma_2 = 60 - 50 = 10 \text{ MPa} \quad \textbf{(Ans.)}$$

From the circle, the counterclockwise angle $2\theta_{p_1}$, measured from CA to the $+\sigma$ axis, is

$$2\theta_{p_1} = 180^\circ - \tan^{-1} \left(\frac{30}{60 - 20} \right) = 143^\circ$$

$$\text{Thus, } \theta_{p_1} = 71.5^\circ$$

The counterclockwise rotation defines the direction of the x' -axis and σ_1 .

Solutions to Assignment #4, cont'd

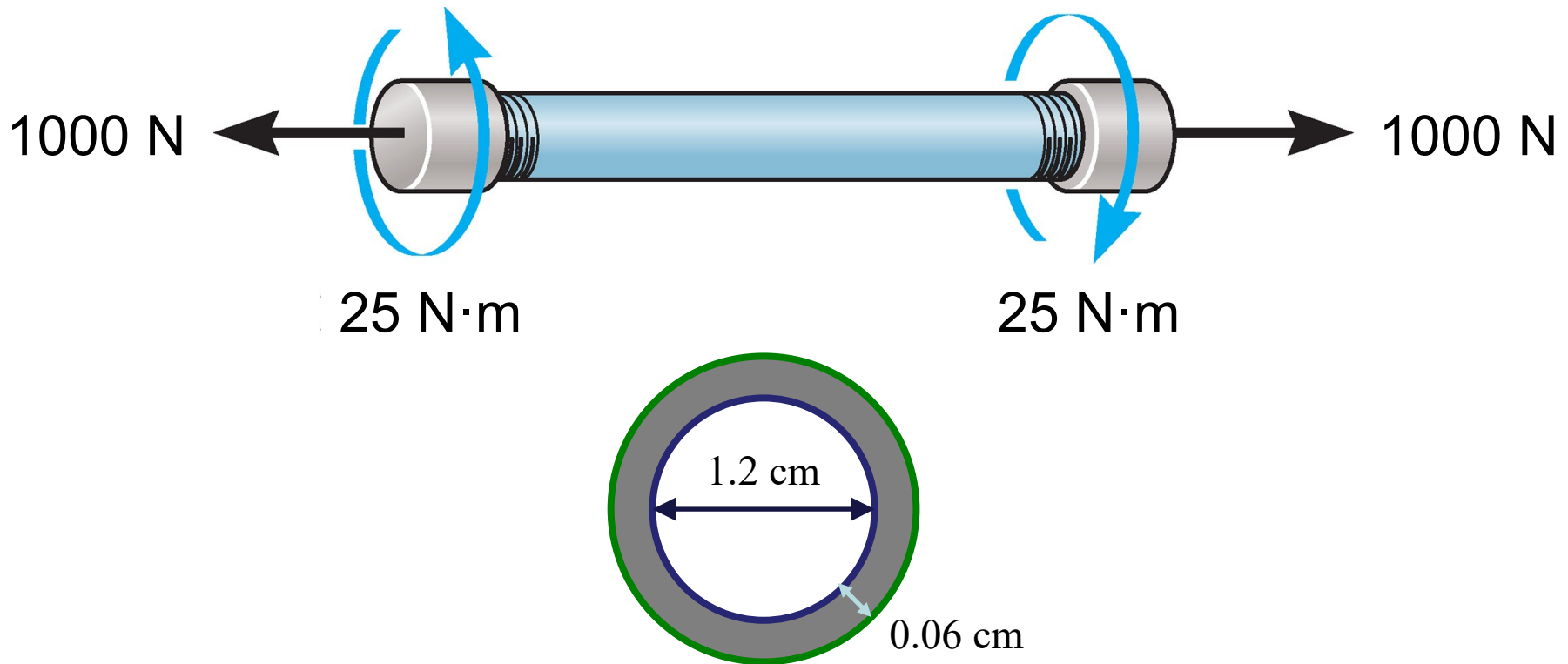
Step 3: Absolute Maximum Shear Stress

Since σ_1 and σ_2 have same sign, applying Eq. (9-13) on page 481 of the textbook we have

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1}{2} = \frac{110}{2} = 55 \text{ MPa} \quad (\text{Ans.})$$

The same results can be obtained by drawing **three** Mohr's circle for each orientation of an element about the x, y, and z axes, and then comparing the maximum shear stress among the three cases (see the figure).

5. The thin-walled pipe has an inner diameter of 1.2 cm and a thickness of 0.06 cm. If it is subjected to an internal pressure of 3.5 MPa and the axial tension and torsional loadings shown, determine the principal stress at a point on the surface of the pipe. (20' points)



Solutions to Assignment #5:

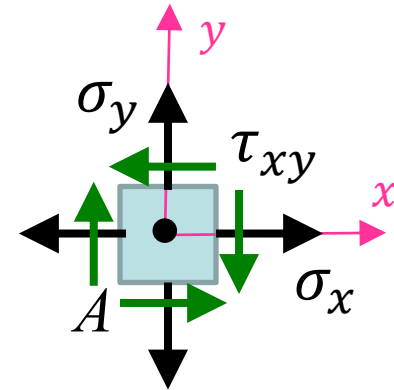
In this gas filled vessel, in addition to the two non-equal normal stresses in the two perpendicular directions (one in the longitudinal direction and the other in the hoop direction), there are shear stress induced by the torque and normal stress induced by the tension force.

Cross-sectional area:

$$A = \frac{\pi}{4} (0.0132^2 - 0.012^2) = 2.375 \times 10^{-5} \text{ m}^2$$

Polar moment of inertia:

$$J = \frac{\pi}{2} (0.0066^4 - 0.006^4) = 9.448 \times 10^{-10} \text{ m}^4$$



$$\sigma_x = \sigma_n + \frac{F}{A} = \frac{pr}{2t} + \frac{F}{A} = \frac{3.5 \times 10^6 \times 0.6}{2 \times 0.06} + \frac{1000}{2.375 \times 10^{-5}} = 59.6 \text{ MPa}$$

$$\sigma_y = \sigma_a = \frac{pr}{t} = \frac{3.5 \times 10^6 \times 0.6}{0.06} = 35 \text{ MPa}$$

$$\tau_{xy} = -\frac{Tc}{J} = -\frac{25 \times 0.0066}{9.448 \times 10^{-10}} = -174.6 \text{ MPa}$$

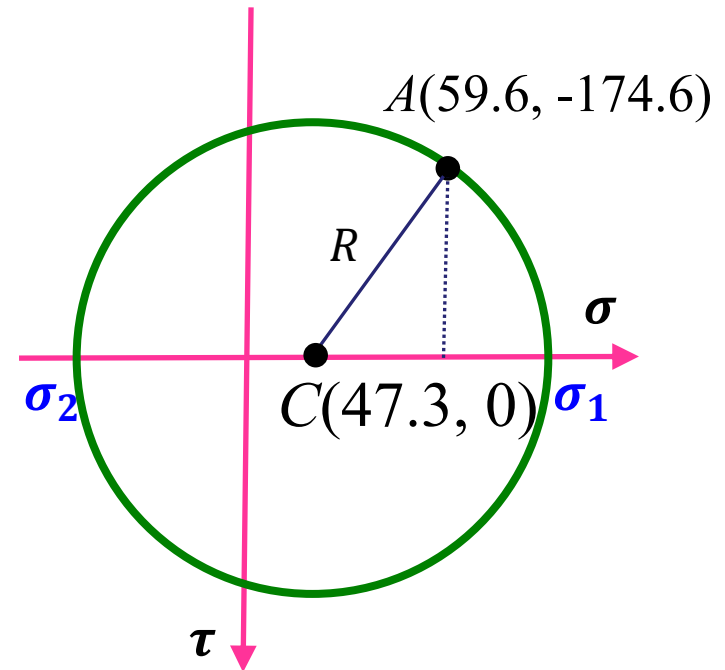
Solutions to Assignment #5, cont'd:

Given $\sigma_x = 59.6$ MPa, $\sigma_y = 35$ MPa, $\tau_{xy} = -174.6$ MPa, the center of the Mohr's circle C is located at the point $C(47.3, 0)$ and the reference point $A(59.6, -174.6)$ are labeled. Then, the radius of the circle is $R = \sqrt{(59.6 - 47.3)^2 + 174.6^2} = 175$, and the circle is constructed.

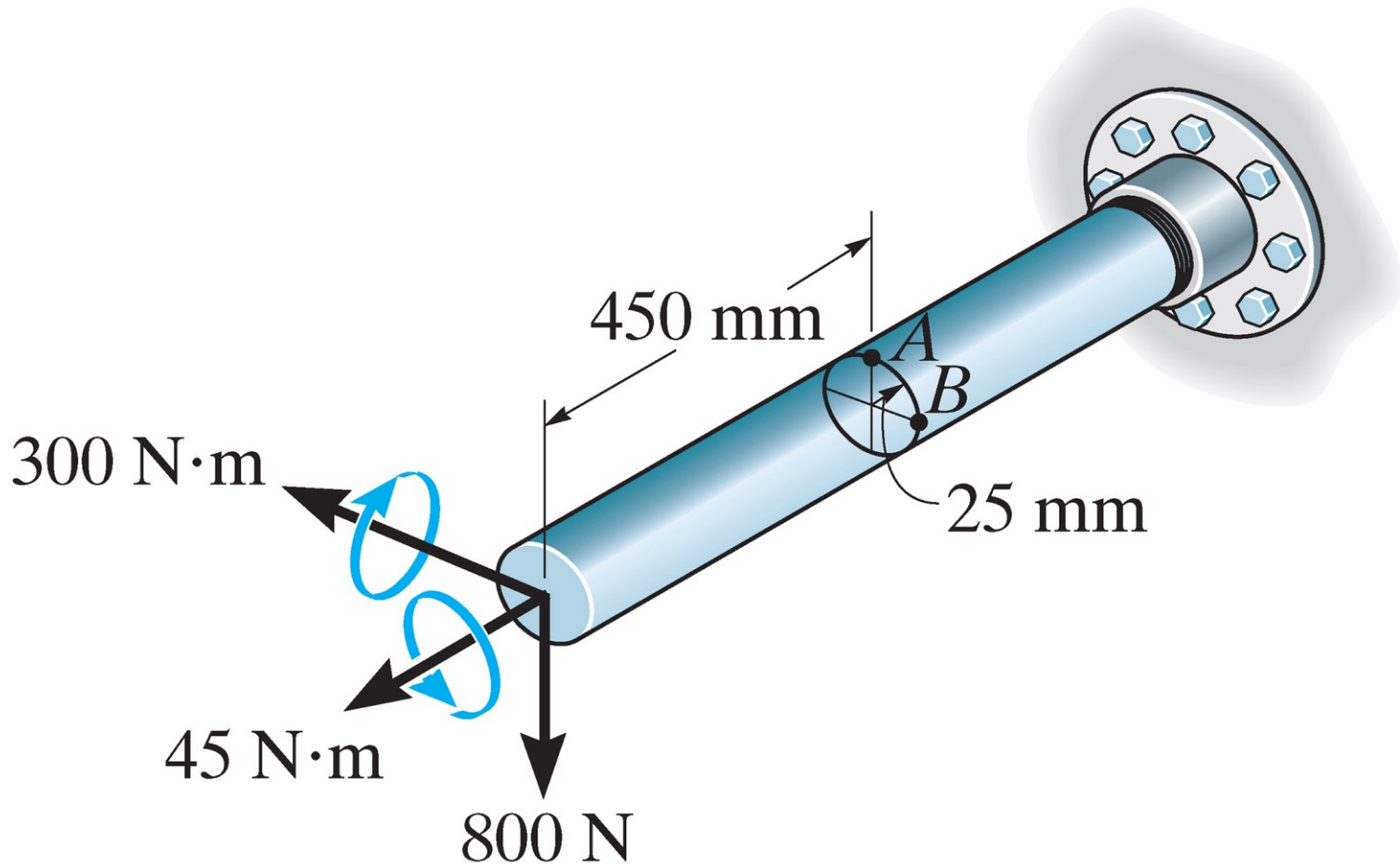
The principal stresses are:

$$\sigma_1 = 47.3 + 175 = 222.3 \text{ MPa} \quad (\text{Ans.})$$

$$\sigma_2 = 47.3 - 175 = -127.7 \text{ MPa} \quad (\text{Ans.})$$



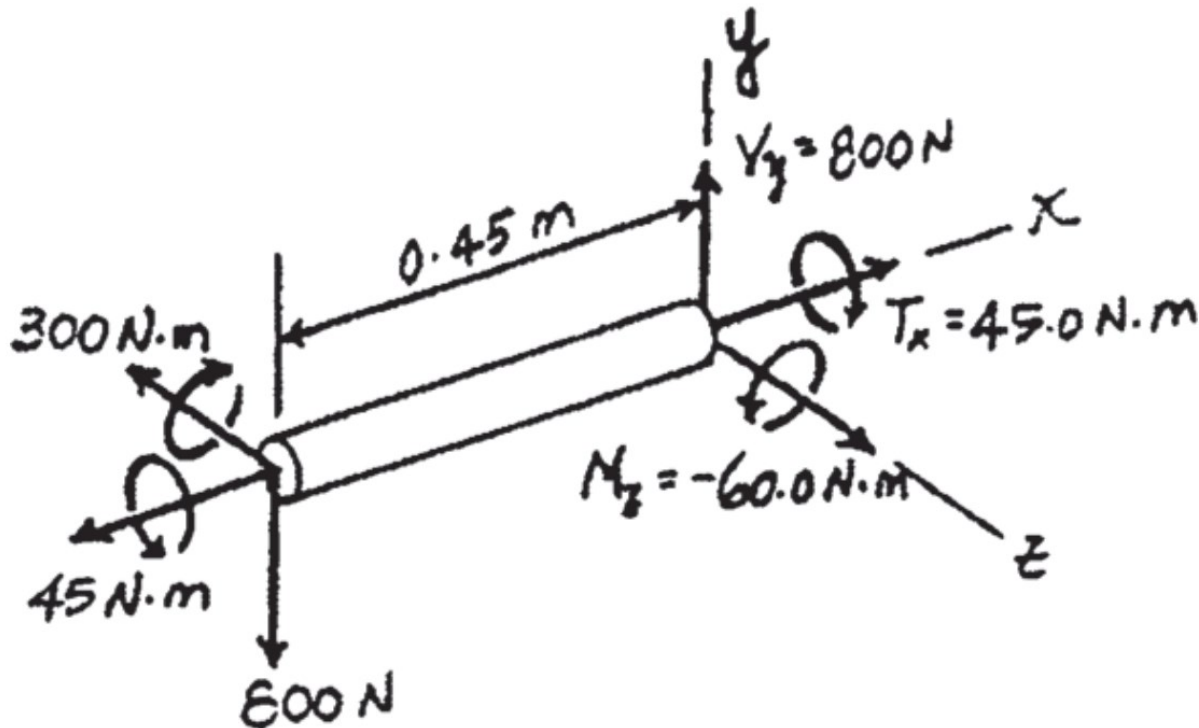
6. The solid shaft is subjected to a torque, bending moment, and shear force. Determine the principal stresses and the absolute maximum shear stress at point A . The radius of the shaft is 25 mm. (25' points)



Solutions to Assignment #6:

Step 1: Internal Forces and Moment

Using the section method through AB plane (perpendicular to the shaft axis) and applying the free body diagram, we get the internal forces and moment as shown in the figure.



Solutions to Assignment #6, cont'd:

Step 2: Section Properties

The moment of inertia with respect to z-axis is

$$I_z = \frac{\pi}{4} \times 0.025^4 = 3.06796 \times 10^{-7} \text{ m}^4$$

The polar moment of inertia of the shaft is

$$J = \frac{\pi}{2} \times 0.025^4 = 6.13592 \times 10^{-7} \text{ m}^4$$

Solutions to Assignment #6, cont'd:

Step 3: Normal Stress and Shear Stress

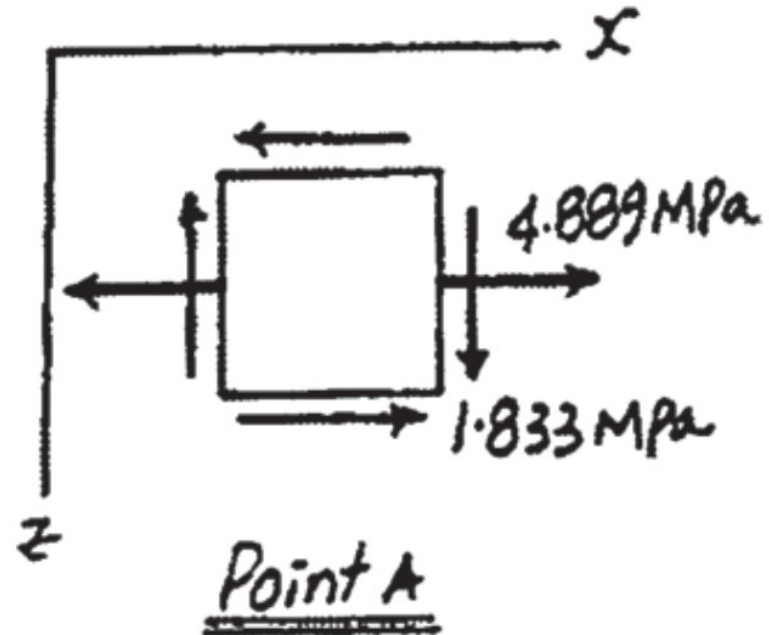
The normal stress at point A can be obtained by applying the flexure formula

$$\sigma_A = -\frac{M_z y}{I_z} = -\frac{-60 \times 0.025}{3.06796 \times 10^{-7}} = 4.889 \text{ MPa}$$

The shear stress at point A can be obtained by applying the torsion formula

$$\tau_A = \frac{Tc}{J} = \frac{45 \times 0.025}{6.13592 \times 10^{-7}} = 1.833 \text{ MPa}$$

The stress state at point A can be drawn as shown in the figure.

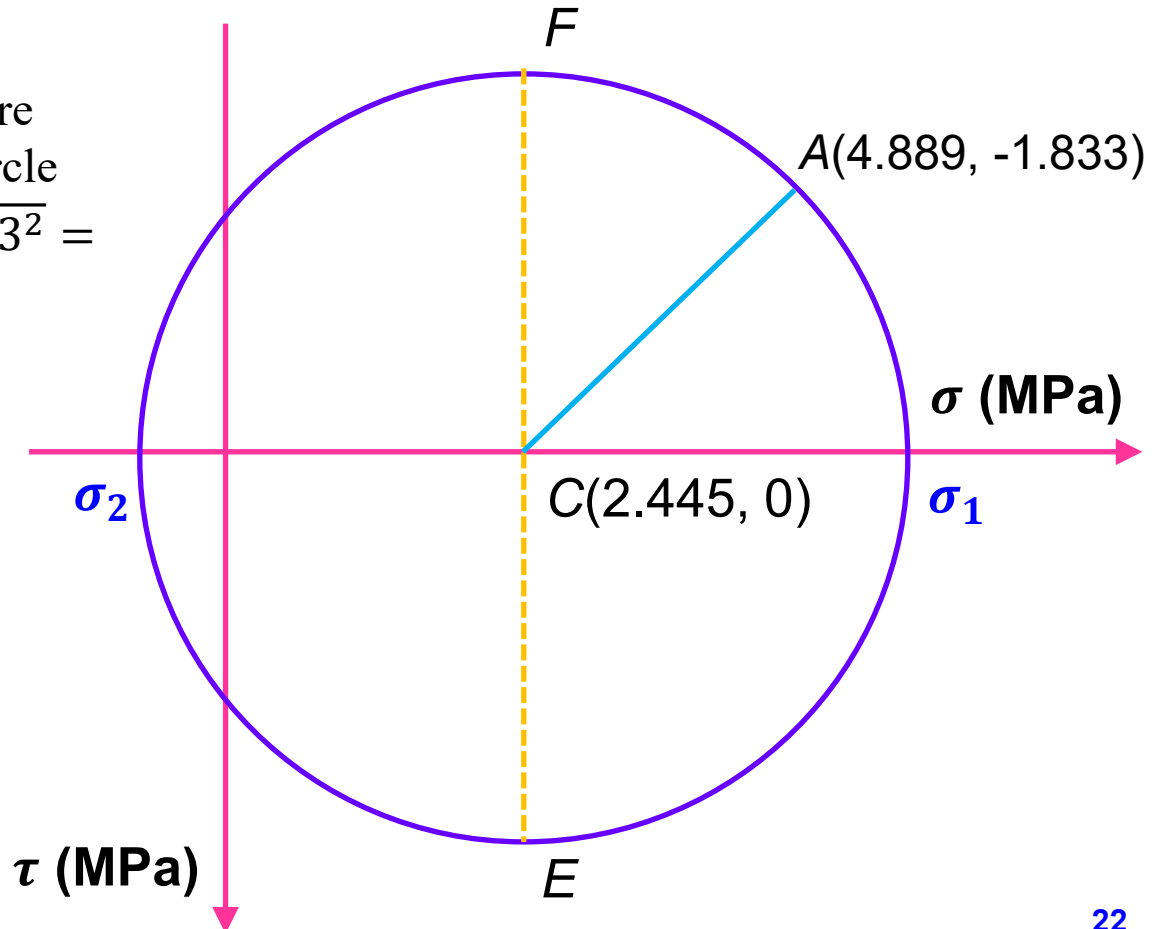


Solutions to Assignment #6, cont'd:

Step 4: Construct the Mohr's Circle

Given $\sigma_x = 4.889$ MPa, $\sigma_z = 0$, $\tau_{xz} = -1.833$ MPa, the center of the circle C is located on the σ -axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_z}{2} = 2.445$ MPa

The point $C(2.445, 0)$ and the reference point $A(4.889, -1.833)$ are labeled. Then, the radius of the circle is $R = \sqrt{(4.889 - 2.445)^2 + 1.833^2} = 3.055$ MPa, and the circle is constructed.



Solutions to Assignment #6, cont'd

Step 5: Principal Stresses

The principal stresses are the two intersections of the Mohr's circle with the σ -axis

$$\sigma_1 = 2.445 + 3.055 = 5.5 \text{ MPa} \quad (\text{Ans.})$$

$$\sigma_2 = 2.445 - 3.055 = -0.61 \text{ MPa} \quad (\text{Ans.})$$

Step 6: Absolute Maximum Shear Stress

Since σ_1 and σ_2 have different sign, the absolute maximum shear stress is:

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = 3.055 \text{ MPa} \quad (\text{Ans.})$$